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	The aim of the research is to compare the development and organization of hemichordates to the development and organization of chordates, thereby to discern the origin of chordate traits in non-chordate ancestors. Hemichordates are the phylum of bilateral animals closest to chordates, the phylum to which vertebrates belong, including humans. Hemichordates have long been thought to possess chordate likenesses and therefore to hold value for identifying chordate traits that were present in ancestors predating chordates. However, the anatomy of these three-part burrowing marine worm-like animals is in many ways unlike chordates, and controversy has surrounded the anatomical comparisons of the past century. Our studies (NASA funding) have entailed the use of modern molecular methods of comparison, particularly in situ hybridization. Thus far, with probes to chordate neural patterning genes, we have assessed hemichordate-chordate similarities of the nervous system. The domains of expression of neural patterning genes have been extensively mapped in the chordate nervous system. At this time, 28 orthologous genes have been used		

Task Description:	from hemichordates, and the conclusion is strongly supported that the anterior-posterior dimension of the hemichordate nervous system is very similar to that of chordates in terms of expression domains of neural patterning genes. This close similarity to chordates was not previously known. The topography of domains is strikingly alike even though the hemichordate nervous system is not centralized as in chordates but is diffuse--a basiepithelial nerve net with tracts of axons along the dorsal and ventral midlines. As a speculation on origins, the domain topography may have arisen in an ancestor with a diffuse nerve net, and the topography was then carried into the chordate nervous system as it was centralized. In the dorsoventral dimension, the hemichordate is more like a protostome (e.g., Drosophila) than a chordate. It has a Bmp-Chordin axis in the Drosophila orientation. Slit and Netrin at the dorsal and ventral midlines respectively pattern axon migration to form the two axon tracts. Our first paper was published last year; another is in preparation. A White Paper prepared by us was accepted, leading to the preparation of a BAC library for S. kowalevskii. A White Paper prepared by us with members of Eric Lander's laboratory has received high priority from NHGRI for sequencing of the entire genome within the next year. The objective of the research is to determine which chordate traits were already present in a non-chordate ancestor shared with hemichordates. The traits include modes of development, domains of expression of genes, molecular interactions, physiology, and morphology. Knowledge about the ancestry of traits, especially molecular and developmental traits dating back to pre-Cambrian ancestors with other phyla, gives insights about what is fundamental and conserved in chordates and what was available to the chordate line for its unique evolutionary innovations. From our studies thus far, the hemichordate seems to have many basic chordate-like characters, but in a less complicated context where processes and components may be easier to elucidate.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Knowledge about the origin of our phylum, the chordates, including ancestry of the body plan, cell types, developmental modes, and molecular components and interactions.
Task Progress:	The importance of the Bmp-Chordin axis in the development of Saccoglossus kowalevskii: We previously reported on the present of a Bmp-Chordin developmental axis in the hemichordate embryo, which was detected by the expression of the bmp2/4 gene on the dorsal midline and the chordin and admp genes on the ventral midline. A similar axis is known to exist in chordates and in Drosophila, suggesting that even the remote pre-Cambrian bilateral ancestor already had such a developmental axis. In this past year, we tested the importance of the axis by altering Bmp levels in the embryo and thereafter scoring the development of the definitive dorsoventral axis of the animal. These functional experiments have worked remarkably well, and we can conclude that every aspect of dorsoventral development depends on this axis in S. kowalevskii. To knockdown Bmp2/4, we used either a dicer mix of siRNAs (Stratagene) for a 900 base pair stretch of S. kowalevskii bmp2/4 coding sequence, or chemically synthesized siRNAs against specific sequences of the bmp2/4 gene. The siRNA preparations were injected into eggs within 5 min after fertilization. Dr. Mark Terasaki (U. Connecticut) helped us greatly with these injections, using an accurate single-shot method, delivering 1% egg volume (0.27 nl for an egg cell of 27 nl). This method gave embryos with very uniform ventralized phenotypes. Over 200 eggs have been injected with full strength dicer material (stock of 50 pg per nl) or defined siRNAs, and all show the same phenotype: a cylindrical symmetric body axis of a ventral type, from which the proboscis falls off on at the time the mouth is forming at the proboscis-collar boundary. The proboscis falls off because the mouth extends across the entire diameter of the body. Flexure of the metasome in the ventral direction fails to occur. The posterior end of the embryo is symmetrical, with a ring of cilia rather than a band of cilia drawn out ventrally on the ventrally bent body. Gill slits, which are normally dorsolateral in location do not form. These are fully ventralized embryos (in press in PLoS Biology). We examined the expression of four genes that normally have ventral domains and are therefore candidates for repression by endogenous Bmp from the dorsal side. Thus, netrin is normally expressed in a thin stripe on the ventral midline. In anti-bmp siRNA injected embryos, netrin expression expands uniformly across the ectoderm, indeed suggesting that Bmp normally represses netrin dorsally. As a second example, pitx expression is altered in the embryos from eggs injected with anti-bmp siRNAs; the posterior ventral domain of pitx expands uniformly throughout the ectoderm whereas the anterior dorsal spot of expression is absent from the detached prosome. These data on pitx from Bmp knockdown embryos complement the results from the Bmp overexpression assays (see below) and suggest strongly that Bmp normally represses the expression of pitx to a posterior ventral stripe in the metasome but activates its expression in the anterior dorsal spot on the proboscis. Third, we examined admp, which is normally expressed in a ventral stripe. In anti-bmp siRNA embryos, admp expression expands uniformly, especially in the proboscis where it becomes very intense. And fourth, in anti-bmp siRNA embryos, dlx expression in the prosome ectoderm remains patchy; however, the dorsal midline expression is absent. Thus, Bmp is likely required for activation of dlx on the dorsal midline but not for prosome ectoderm expression. To achieve the opposite condition of ubiquitous overexpression of Bmp2/4, we added zebrafish recombinant Bmp 4 protein to seawater. All treatments significantly altered development to produce fully dorsalized embryos. Treated embryos are cylindrical in shape, more elongated than normal, flattened at the proboscis tip, and lacking the dorsal flexure that normally results from asymmetric tissue extension. The mouth is absent, and instead of a thin neck, the mesosome is thickly connected to the prosome. Also the first gill slits are entirely lacking, although at very low levels of Bmp exposure (10ng/ml) a deep circumferential groove forms in the ectoderm at the position along the anteroposterior axis at which the gill slits would first perforate. The organization of the anteroposterior axis is generally unaffected. We then assessed the effects ubiquitous Bmp on dorsoventral gene expression in the three germ layers. First we assayed the expression of the bmp2/4 gene itself and found that its mRNA is elevated to detectable levels throughout the ectoderm rather than remaining localized to the dorsal midline as in control embryos, implying an autoactivating circuitry. Then we assayed six genes that, in normal embryos, are expressed on or near the dorsal midline in either the ectoderm or endoderm, as candidates for activation by Bmp. The expressions of dlx, tbx2/3, pitx, nkx2.3/2.5, hex, and pax1/9 all expanded greatly to encircle the body, indicating their activation by Bmp. The first three of these are expressed in ectoderm and the last three in endoderm and mesoderm. Thus, we think all three germ layers are patterned by Bmp. As a final case, we chose admp as a candidate for Bmp repression since this gene is normally expressed only on the ventral midline of the ectoderm and endoderm. In Bmp-treated embryos, admp was greatly reduced, leaving a residual small spot in the anterior endoderm. Thus, uniform exposure of the embryo to excess Bmp protein activates the widespread expression of those genes normally expressed dorsally, and represses those normally expressed ventrally.

These findings indicate that dorsal midline Bmp serves in normal dorsoventral development to pattern all the gene expression domains and anatomical/physiological specializations that we have examined on all three germ layers. It is the basis of the entire dorsoventral dimension of the animal. There are two interesting exceptions. In Drosophila and vertebrates, Bmp represses neurogenesis in part of the ectoderm, and this is an important step in centralization of the nervous system. However, in hemichordates we expect no such repression since neurogenic and epidermal territories are not segregated. To test whether Bmp has no repressive effect on neurogenesis, we examined the expression of elv, a pan neural gene, in Bmp protein-exposed embryos, and found that its expression is not down-regulated. It is expressed at uniform high levels around the embryo's circumference and is similar to controls along the anteroposterior axis. Therefore, unlike insects and vertebrates, hemichordates do not have a default gene regulatory circuit by which Bmp represses neural fates during early development. This is a significant difference form chordates. We think the evolution of such circuitry was an important step in the chordate lineage. Finally, the mouth of the hemichordate is on the Chordin (low Bmp) side of the embryo, whereas in chordates it is on the Bmp side. To us, this difference suggests that the mouth of the deuterostome ancestor moved to the opposite side in the chordate lineage.

In summary, it appears from our studies of hemichordates that chordate evolution entailed little change in the anteroposterior dimension of the domain map or in the four anteroposterior signaling centers of the deuterostome ancestor (reported last year). In the dorsoventral dimension, the Bmp-Chordin axis was already well elaborated in the ancestor, with many domains and anatomical differentiations were already located on it, including the gill slits and post-anal tail, and these were carried forward in the both the chordate and hemichordate lineages. The greatest changes seem to have occurred in the dorsoventral dimension with regard to: 1) the centralization of the nervous system along the Bmp-Chordin axis, which involved a novel separation of neural (low Bmp) and epidermal (high Bmp) territories; b) the invention of the notochord and prechordal endomesoderm as elongating signaling centers releasing Bmp antagonists and Wnt antagonists; and c) the relocation of the mouth toward the high Bmp side.

Bibliography Type:	Description: (Last Updated: 04/02/2007)
Articles in Peer-reviewed Journals	Gerhart J, Lowe C, Kirschner M. "Hemichordates and the origin of chordates ." Curr Opin Genet Dev. 2005 Aug;15(4):461-7. PMID: 15964754 , Aug-2005
Articles in Peer-reviewed Journals	Lowe C J, Terasaki M, Wu M, Freeman R, Runft L, Kwan K, Haigo S, Aronowicz J, Lander E, Gruber C, Smith M, Kirschner M, Gerhart J. "Dorsoventral patterning in hemichordates: insights into early chordate evolution." PLoS Biol. 2006 Sep;4(9):e291. PMID: 16933975 , Sep-2006
Articles in Peer-reviewed Journals	Lowe CJ, Tagawa K, Humphreys T, Kirschner M, Gerhart J. "Hemichordate embryos: procurement, culture, and basic methods." Methods Cell Biol. 2004;74:171-94. Review. PMID: 15575607 , Sep-2004
Articles in Peer-reviewed Journals	Lowe CJ, Wu M, Salic A, Evans L, Lander E, Stange-Thomann N, Gruber CE, Gerhart J, Kirschner M. "Anteroposterior patterning in hemichordates and the origins of the chordate nervous system." Cell. 2003 Jun 27;113(7):853-65. PMID: 12837244 , Jun-2003